

Lab 1: Branch Predictor Design and Analysis

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Warmup Task: Baseline(Bimodal Predictor) Run

The mean MPKI is 11.0887, and the mean CycWPKI is 281.0110

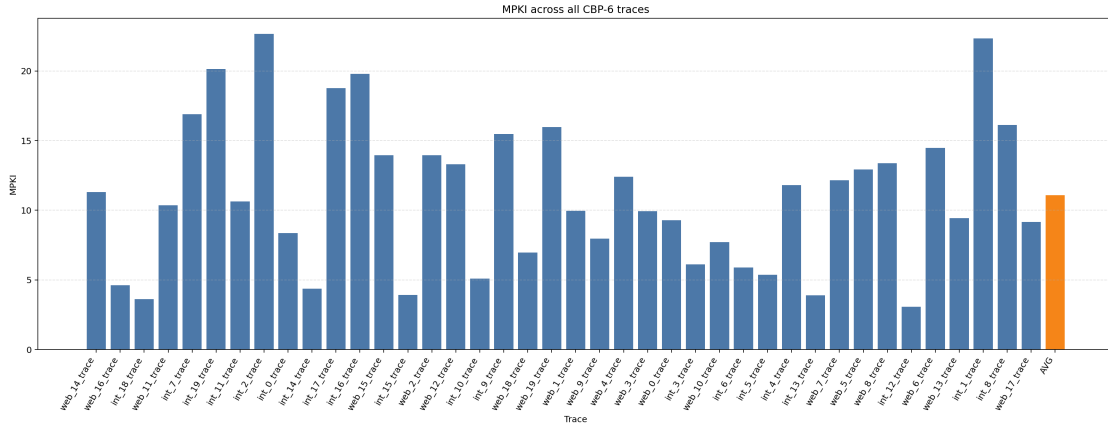


Figure 1: MPKI across all 40 traces with bimodal predictor.

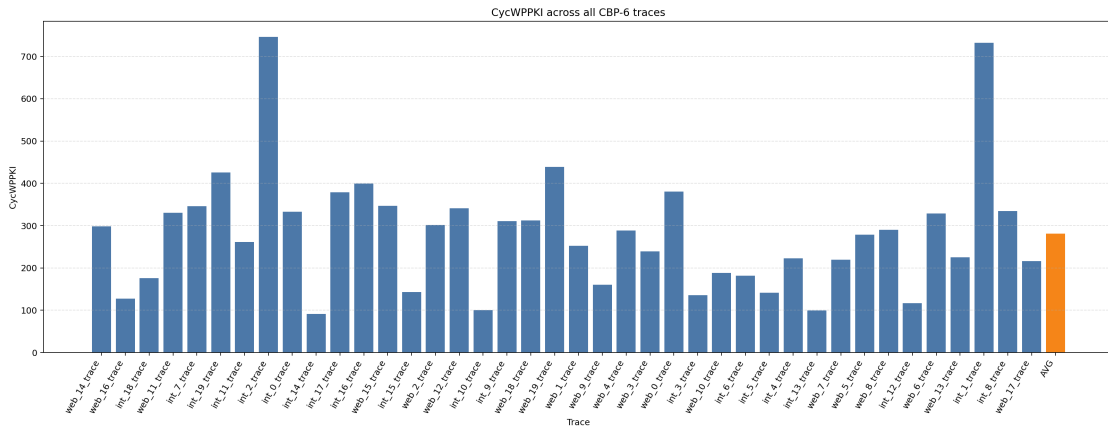


Figure 2: CycWPKI across all 40 traces with bimodal predictor.

Task 1: Local-History Predictor (PAg)

The PAg predictor's mean MPKI is 11.6784, and its mean CycWPPKI is 286.3173

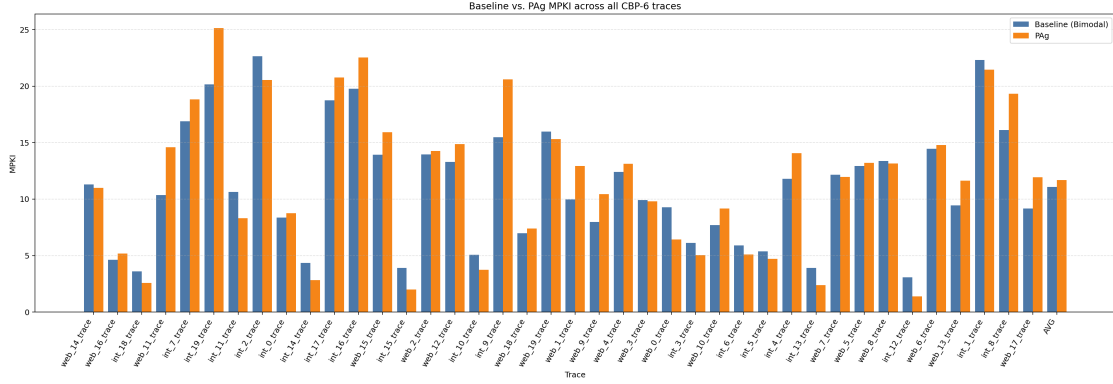


Figure 3: Baseline and PAg MPKI across all 40 traces

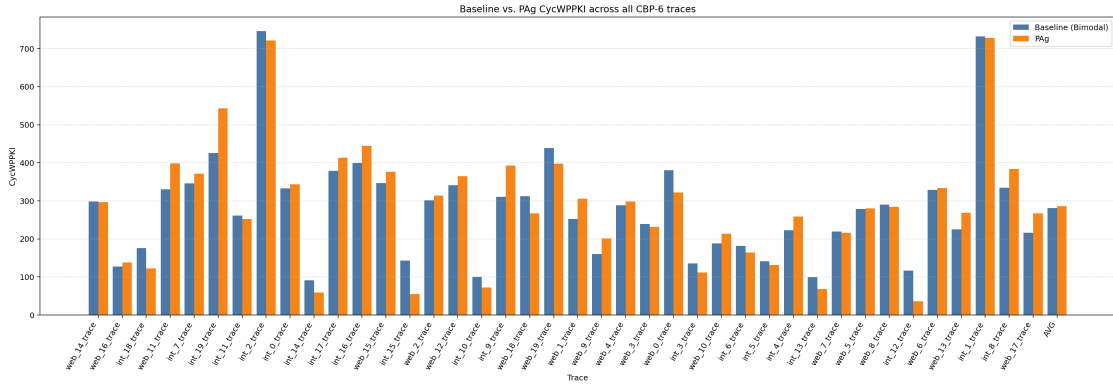


Figure 4: Baseline and PAg CycWPPKI across all 40 traces

Task 2: Tournament Predictor

The tournament predictor's mean MPKI is 10.8785, and its mean CycWPPKI is 271.5273

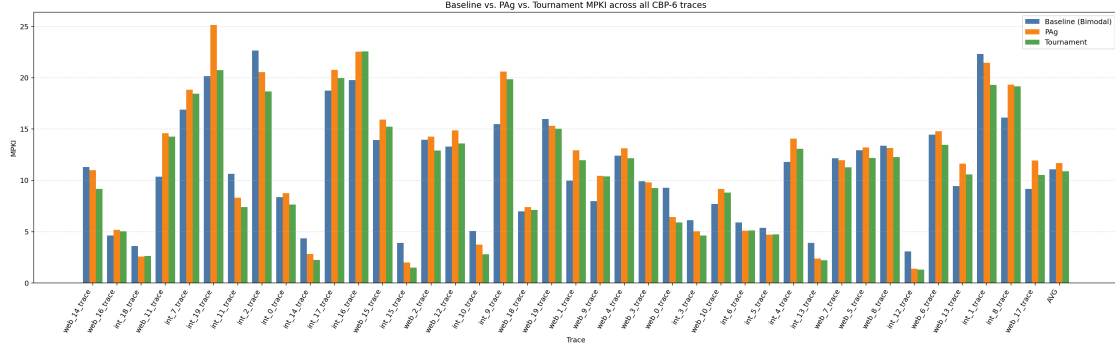


Figure 5: Baseline, PAg, and tournament MPKI across all 40 traces

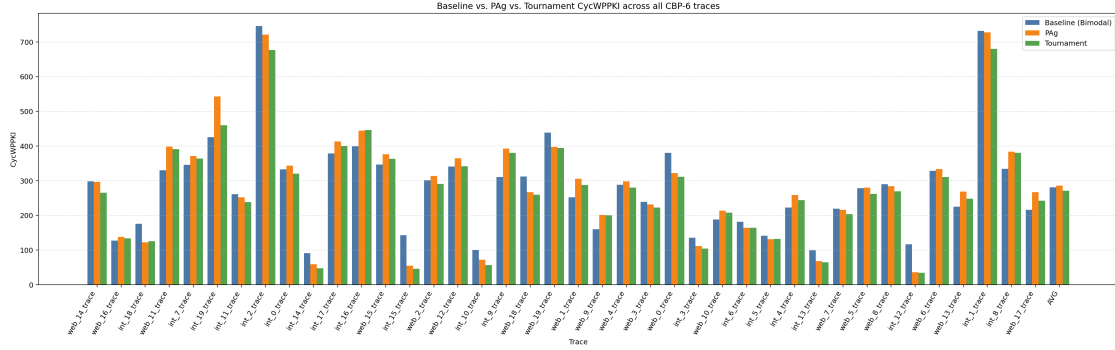


Figure 6: Baseline, PAg, and tournament CycWPPKI across all 40 traces

Task 3: Hashed Perceptron Predictor

The hashed perceptron predictor's mean MPKI is 9.3776, and its mean CycWPPKI is 238.4978

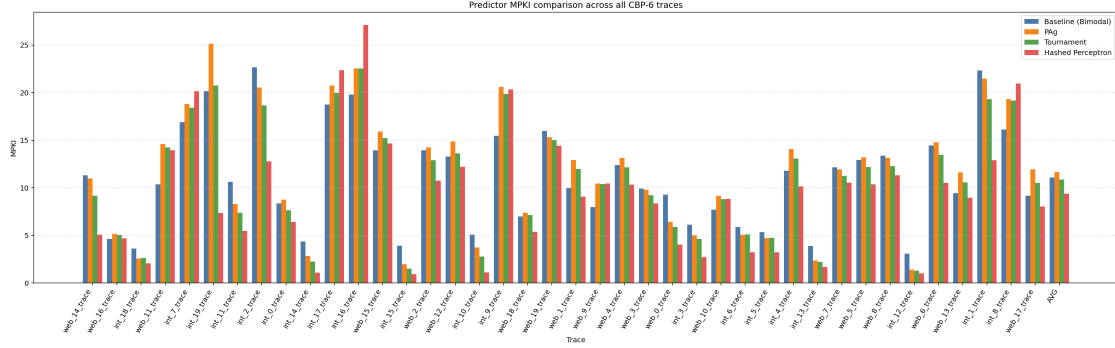


Figure 7: Baseline, PAg, tournament, and hashed perceptron MPKI across all 40 traces

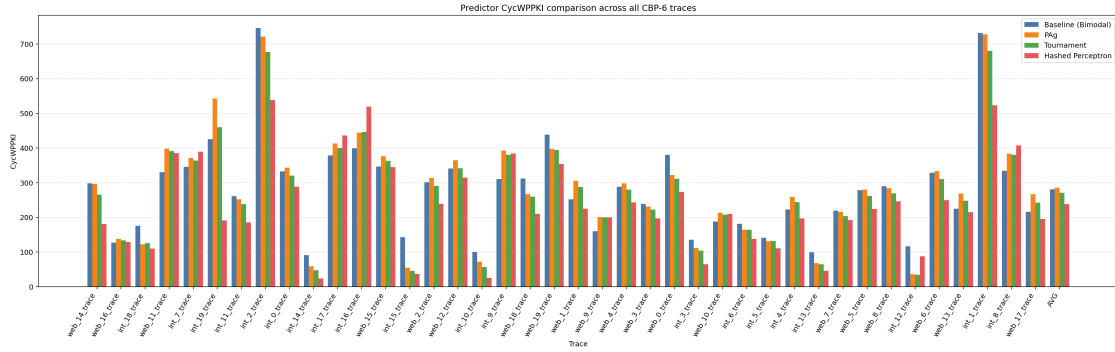


Figure 8: Baseline, PAg, tournament, and hashed perceptron CycWPPKI across all 40 traces